

- In scenarios where single board computers can be used, JavaScript or Node.js code is executed in these devices without any issues.

JavaScript is suitable for devices because of its ability to react to events and asynchronous code execution. It could be a great option for quick prototyping as well. The ability of JavaScript to provide a uniform interface across heterogeneous devices is also a key factor in choosing it for IoT.

There are plenty of JavaScript based frameworks/libraries available for IoT and other related domains (Figure 2).

This article introduces four of these tools, which are listed below:

- Favor
- JerryScript
- IoT.js
- Cydon.js

Favor

Favor is a simple and powerful library for IoT, which hides the complex and inconsistent interfaces provided by heterogeneous devices, from the app developer. The single API provided to access these devices makes it simple for the developer to control various devices.

As narrated in the official documentation (<https://github.com/favor/it>), if you wish to have a function like reading temperature from various heterogeneous devices without considering their hardware differences, using commands

similar to what's shown below, then Favor is the library that will assist you.

```
$$('temperature').get(function(temp) {
    console.log('The temperature is ', temp)
});
```

The Favor documentation uses the term *hardware agnostic*, which implies that developers can write the business logic of the app without thinking about the actual hardware in which the code is going to get executed.

Presently, Favor supports devices of Linux origin such as Raspberry Pi, Beaglebone, etc. However, any Linux device with the ability to run Node.js can be used.

Favor is able to achieve the hardware agnostic feature by maintaining a configuration file. This consists of details describing the structure of the hardware. Favor queries this configuration file when communicating with the actual device. This functionality could be compared with how jQuery parses and interacts with the DOM of a Web page. Once the parsing is done, then jQuery is capable of accessing any DOM element with the simple \$ functionality. Similarly, Favor is able to exhibit jQuery style functionality in accessing the devices. As narrated earlier, any device with a temperature sensor will interact with the code segment shown below:

```
$$('temperature').get(callback)
```

The advantages of using Favor are:

- Uniform and simple API for communicating across various devices/protocols.
- Clear demarcation between hardware and software, which increases the modularity of the application. It separates the business logic from the hardware.
- The 'Write Once - Run Anywhere' feature makes it more suitable for a variety of application scenarios.

Favor can be installed easily with Node.js, using the following command:

```
npm install favor
```

A simple but effective demo of a sample application with Favor is available at <https://www.youtube.com/watch?v=ujHa-I3ZRUM>.

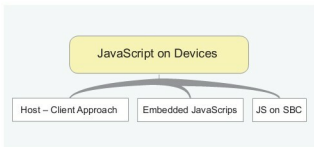


Figure 1: JavaScript on devices

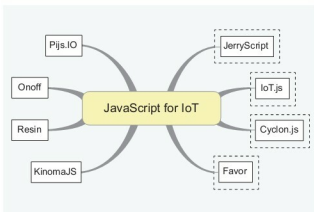


Figure 2: JavaScript for IoT

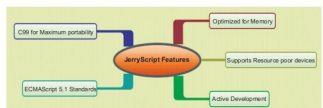


Figure 3: JerryScript features